

Annexure-1: Detailed Lecture-wise Breakup of Course Content

Lecture No. (1 hour each)	Content
	Introduction to Groundwater:
1	Definitions of groundwater and associated terminologies, groundwater and water cycle
2	Importance of groundwater, Indian and global scenarios of groundwater, salient groundwater issues in India.
	Occurrence of Groundwater:
3, 4	Hydrologic zones below the ground and their hydraulic characteristics, types of subsurface formations, aquifer and confining layers
5, 6, 7, 8	Origin and age of groundwater, groundwater regions of India, types of aquifers and their physical and hydraulic properties
9, 10	Water-yielding mechanisms of aquifers, aquifer heterogeneity and anisotropy.
	Principles of Groundwater Flow:
11, 12	Concepts of fluid potential and hydraulic head, Darcy's Law and its validity, specific discharge and seepage velocity, concept of transmissivity
13, 14, 15, 16	Overview of springs and anthropogenic impacts, governing equations of groundwater flow in confined, leaky and unconfined aquifers, solutions of groundwater-flow equations
17, 18, 19	Flow net and its utility, refraction of groundwater-flow lines, analyses of steady groundwater flow in confined and unconfined aquifers.
	Impacts of Environmental Factors on Groundwater:
20, 21	Natural and anthropogenic factors affecting groundwater, Temporal variations of groundwater levels, streamflow versus groundwater levels
22, 23	Impacts of natural factors and human developmental activities on groundwater such as precipitation, evapotranspiration, floods, droughts, cyclones, earthquakes, tsunami, volcanic eruptions, pumping, agriculture, urbanization, industrialization, mining, dumping of solid and liquid wastes, etc.
	Groundwater Investigation:
24, 25	Scale of groundwater investigation, groundwater exploration methods and their limitations, construction of well logs and geologic profiles, stratigraphy analysis
26, 27, 28	Monitoring of groundwater level and quality, laboratory and field methods for determining aquifer parameters, field investigation for estimating potential recharge.
	Estimation of Recharge and Groundwater Resource:
29, 30, 31	Definition of recharge and its types, methods for estimating groundwater recharge and their limitations
32, 33	Quantitative and qualitative assessment of groundwater resources, evaluation of groundwater-extraction level.
	Construction of Water Wells:
34, 35	Types of wells and their functions, classification of water wells, design of water supply wells, methods for constructing shallow pumping wells
36, 37, 38, 39	Methods for drilling deep pumping wells, well casings and screens, gravel packing, cementing, methods of well development, determination of well

	yield, protection of wells from contamination, infiltration gallery and radial collector wells.
	Well Hydraulics and Multiple Well Systems:
40, 41, 42	Basics of well hydraulics, analysis of steady flow to pumping wells in confined and unconfined aquifers, hydraulics of partially penetrating wells
43, 44, 45	Evaluation of well losses, well specific capacity and well efficiency, analysis of well interference (confined and unconfined aquifers) and its applications, analysis of multiple well systems.
	Groundwater Management:
46, 47, 48	Advantages of aquifer storage, groundwater budget, management potential of aquifers, dynamic equilibrium in aquifers and impacts of excessive pumping
49, 50, 51	Paradox of safe aquifer yield, concepts of groundwater resilience and sustainable aquifer yield, stream-aquifer interaction, concept of conjunctive use and its advantages, case studies
52, 53, 54	Cost-effective and sophisticated techniques for groundwater conservation and augmentation, measures for protecting groundwater quality, case studies
55, 56	Roles of geospatial, IoT and AI/ML techniques in sustainable management of water resources, case studies.
Project presentation will be within class time slots.	